

IEOR 4404 – Fall 2025

Practice Midterm (Structural Variant B)

Problem 1 (Discrete ITM + A/R on a zero-truncated Poisson). Let X follow a *zero-truncated* Poisson distribution,

$$\mathbb{P}(X = i) = \frac{e^{-\lambda} \lambda^i / i!}{1 - e^{-\lambda}}, \quad i = 1, 2, \dots, \quad \lambda > 0.$$

- (a) Derive an ITM sampler by using the recursion $\mathbb{P}(X = i + 1) = \frac{\lambda}{i+1} \mathbb{P}(X = i)$, $i \geq 1$, and an initialization at $i = 1$.
- (b) Show that the algorithm “sample $Y \sim \text{Pois}(\lambda)$; if $Y \geq 1$ accept and set $X = Y$; else repeat” is an acceptance–rejection method. Specify the proposal g , the target f , the constant C , and the acceptance probability.

Problem 2 (Continuous ITM + tail check for A/R). Consider the *Lomax* (Pareto II) distribution on $(0, \infty)$ with cdf

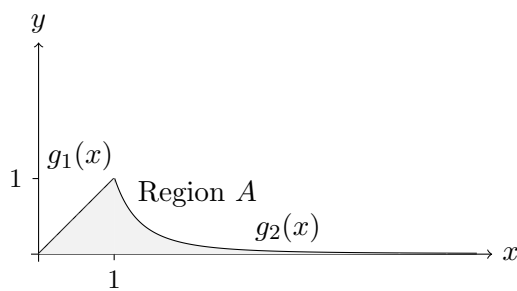
$$F(x) = 1 - (1 + \theta x)^{-\alpha}, \quad \alpha, \theta > 0.$$

- (a) Derive an ITM sampler for X by inverting F .
- (b) Suppose you propose from the same family with parameters $(\tilde{\alpha}, \theta)$, i.e., heavier or lighter tails depending on $\tilde{\alpha}$. Determine precisely for which relationship between α and $\tilde{\alpha}$ the ratio $f(x)/g(x)$ is bounded on $(0, \infty)$, and conclude when the A/R scheme is valid and when it fails.

Problem 3 (Uniform points in a piecewise region via ITM). Let A be the region under g_1 on $[0, 1]$ and under g_2 on $[1, \infty)$:

$$g_1(x) = x \cdot \mathbf{1}\{0 < x \leq 1\}, \quad g_2(x) = x^{-3} \cdot \mathbf{1}\{1 \leq x < \infty\}.$$

- (a) Compute the total area of A and the marginal cdf of X for a point (X, Y) dropped uniformly in A .
- (b) Give an ITM sampler for X , then conditionally generate $Y \mid X$ uniformly on $[0, g(X)]$.



Problem 4 (Composition method with a different kernel). Consider the nonnegative random variable X with density

$$f_X(x) = \int_0^\infty \frac{4t}{(1+xt)^5} e^{-t} dt, \quad x \geq 0.$$

Derive a simulation algorithm by viewing the integrand as a mixture: first generate $Z \sim \text{Exp}(1)$, then generate $X \mid Z = z$ by inverting the conditional cdf corresponding to the kernel $\frac{4z}{(1+xz)^5}$.